# Supervyse OPF Intel Quick Start Guide

***Revision 1.0***

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# Revision History

| Revision Number | Date | Author | Reviewer | Summary |
|---|---|---|---|---|
| 1.0 | 2025-02-27 | Youmin Chen | | RV24.12.3 Release Document |
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# Introduction

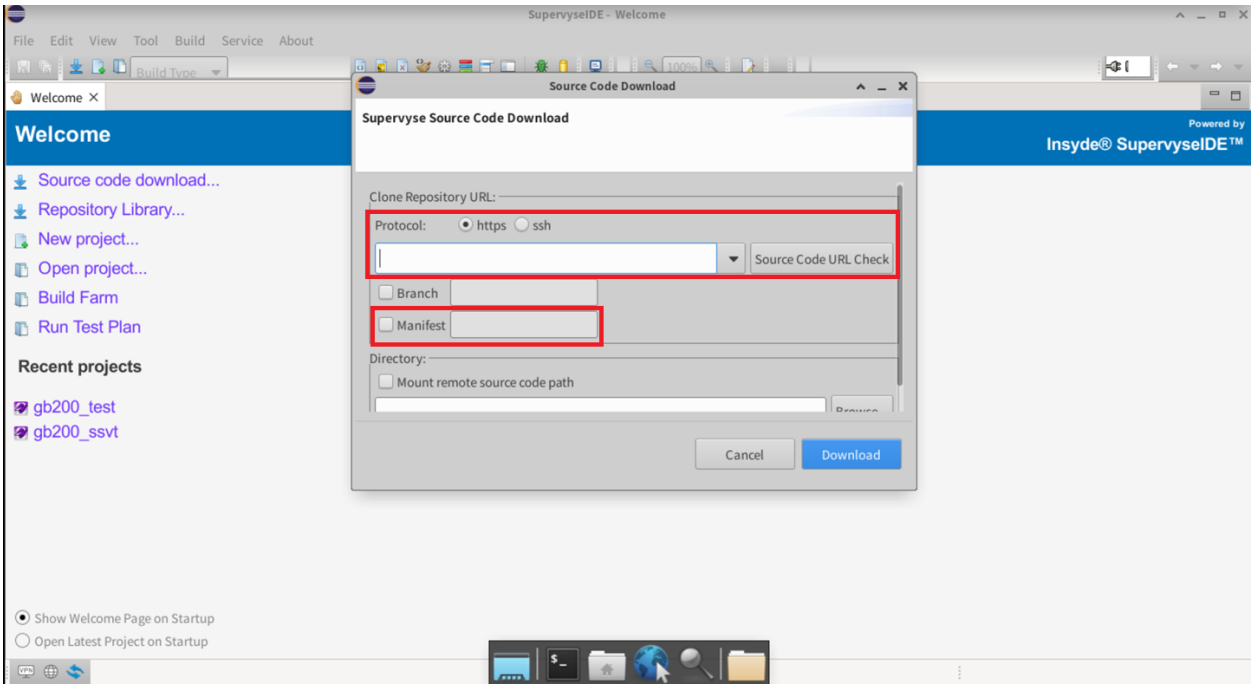
This page introduces how to setup a build environment for compiling Supervyse OPF code, including the software packages you need to install.

# Download SDK from IDE

Supervyse OPF is a Larger-scale project with many git repositories. Insyde uses Google repo tool to manage Supervyse OPF. Users can directly use IDE to get the required code.

## Use IDE to Download SDK Guide

Please refer to the **Download Source Code** section in the **SupervyseIDE_OPF_User_Guide.pdf** file included in the SupervyseIDE installation package.




## Download OPF Intel SDK

In the IDE Download Source Code window, you can select the download url:

https://bitbucket.org/insydesw/g4_manifest-intel

| Manifest xml file | Description |
|---|---|
| AST2600_OPF-BirchStream_master.xml | The manifest xml file is used for download of development branch. It will be updated to the latest status and stability cannot be guaranteed. |
| AST2600_OPF-BirchStream_RV24.06.4.xml | The manifest xml file is used for downloading of tag RV24.06.4.xml |
| AST2600_OPF-BirchStream_RV24.12.3.xml | The manifest xml file is used for downloading of tag RV24.12.3.xml |

Refer to the table above to select the manifest you need:

AST2600_OPF-BirchStream_master.xml

AST2600_OPF-BirchStream_RV24.06.4.xml

AST2600_OPF-BirchStream_RV24.12.3.xml

# Setup of the Build Environment

Before using the SDK, you should consider what your needs/requirements will be.
Supported Operating System list:

- Ubuntu 18.04/64 (EOS in June 2023)
- Ubuntu 20.04/64
- Ubuntu 22.04/64

## Install the packages on your Ubuntu 18.04/64 & 20.04/64

Please install the following tools.

```
$ sudo apt-get update
$ sudo apt-get install -y git build-essential libsdl1.2-dev texinfo gawk chrpath diffstat zstd pigz
$ sudo apt-get install -y unzip gcc-multilib g++-multilib xxd curl liblz4-tool
$ sudo apt-get install -y python jq python3-pip
$ sudo -H pip3 install yq
```

## Install the packages on your Ubuntu 22.04/64

Please install the following tools.

```
$ sudo apt-get update
$ sudo apt-get install -y git build-essential libsdl1.2-dev texinfo gawk chrpath diffstat zstd pigz
$ sudo apt-get install -y unzip gcc-multilib g++-multilib xxd curl liblz4-tool
$ sudo apt-get install -y jq python3-pip
$ sudo -H pip3 install yq
```

## Link /bin/sh to bash

You must use bash instead of sh to build the code

```
$ sudo dpkg-reconfigure -plow dash
```

Choose '**No**' in the 'Configuring dash' popup dialog and press enter.

If it doesn't work, try:

```
$ sudo rm /bin/sh

$ sudo ln -s /bin/bash /bin/sh
```

- To ensure that the link was changed to bash:

```
$ ls -l /bin/sh
```

# Build Source Code

## BirchStream Platform

### 1.1.1 BirchStream Platform

```
$ source setup  intel-bhs-insyde
$ bitbake intel-platforms
```

The output image with time stamp, " intel-platforms-intel-bhs-insyde-xxxxxxxxxxxxxx.auto.mtd, will be placed in:

**<SDK>/build/intel-bhs-insyde/tmp/deploy/images/intel-bhs-insyde/**

To find the latest image quickly, the latest output image is linked to the following symbolic link:

**<SDK>/build/intel-bhs-insyde/tmp/deploy/images/intel-bhs-insyde/image-mtd**

# Signing BMC Firmware

## With Default Private Key

- OPF will use the following default private key to sign BMC firmware.
- The private key is used to sign the image.

<SDK>/meta-phosphor/recipes-phosphor/flash/files/OpenBMC.priv

- The publickey corresponding to private key is in the following location.
- The publickey is used to verify whether the image is valid or not when firmware update is procceeding.

**<SDK>/build/tmp/work/$MACHINE-openbmc-linux-gnueabi/$RECIPE/$VERSION/rootfs/etc/activationdata/key/publickey**

**EX: (AST26x0)**

**<SDK>/build/tmp/work/evb_ast26x0_insyde-openbmc-linux-gnueabi/obmc-phosphor-image/1.0-r0/rootfs/etc/activationdata/OpenBMC/publickey**

- If you decompress the BMC firmware image, there are some .sig files which are used to be verified with publickey

    1. image-full.sig
    2. image-kernel
    3. image-kernel.sig
    4. image-rofs
    5. image-rofs.sig
    6. image-runtime
    7. image-runtime.sig
    8. image-rwfs
    9. image-rwfs.sig
    10. image-u-boot
    11. image-u-boot.sig
    12. MANIFEST
    13. MANIFEST.sig

## With Specific Private Key

- If you want to use other private key, not default one. You can use following way to produce your own private key.

:* Generate RSA1024 private key named "OpenBMC.priv"

```
# openssl genrsa -out OpenBMC.priv 1024
```

:* Generate RSA2048 private key named "OpenBMC.priv"

```
# openssl genrsa -out OpenBMC.priv 2048
```

:* Generate RSA4096 private key named "OpenBMC.priv"

```
# openssl genrsa -out OpenBMC.priv 4096
```

- Export some environment variables
- Build firmware image with environment variables

```
# export BB_ENV_EXTRAWHITE="$BB_ENV_EXTRAWHITE SIGNING_KEY"
# SIGNING_KEY=/path/to/secure/OpenBMC.priv bitbake obmc-phosphor-image (or intel-platforms)
```

- **It's a normal behavior when BMC Firmware update failed after private key has been changed.**

# Build performance

- By default, Yocto use all CPU of the build machine to do the jobs.

If user want to specify the number of the jobs it can perform,

add following options to /conf/local.conf

```
BB_NUMBER_THREADS = "4"
PARALLEL_MAKE = "-j 4"
```

- Yocto uses build/downloads and build/sstate-cache as your default directories.

You can change them by adding the following options to build/conf/local.conf.

This will increase speed when you rebuild the same project with the same settings.

```
BB_GENERATE_MIRROR_TARBALLS = "1"
DL_DIR="/where/to/your/downloads"
SSTATE_DIR="/where/to/your/sstate-caches"
```

# Troubleshooting

It is suggested that you double-check your versions of both xq and yq before you start to build OPF.

On Ubuntu 18.04:

```
$ xq --version
xq 2.12.0
$ yq --version
yq 2.12.0
```

On Ubuntu 20.04:

```
$ xq --version
xq 3.2.1
$ yq --version
yq 3.2.1
```

On Ubuntu 22.04:

```
$ xq --version
xq 3.2.2
$ yq --version
yq 3.2.2
```

Try upgrading your xq or yq if their versions are much lower than above. You may need to uninstall your local versions first.

```
$ pip3 uninstall xq
$ pip3 uninstall yq
$ sudo -H pip3 install yq
```

Then, exit and login again to take effect.